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Effects of Cell Density and Temperature on Oxygen Consumption Rate for Different Mammalian Cell Lines

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Abstract: Oxygen consumption rates were measured in a respirometer for different mammalian cell lines (BHK, murine hybridoma, and CHO), and the effects of cell density (1–20 million cells/mL) and temperature (6 to 37°C) on specific oxygen consumption rate were investigated. The specific oxygen consumption was cell line dependent. For a given temperature, the murine hybridoma cells had the lowest and the CHO cells had the highest oxygen consumption rate. The specific oxygen consumption rate was not affected by the cell concentration for cell densities between 1 and 20 million cells/mL. However, artificial trends implicating the effects of cell density were obtained when traditional analysis was used and the probe response time was neglected. A detailed mathematical analysis was presented to investigate the magnitude of errors originating from neglecting the probe response time for the calculation of oxygen consumption rate. The error was significant, especially when the probe response was slow and/or the oxygen consumption was fast. Temperature influenced the specific oxygen consumption rate similarly for the cells studied, and about 10% decrease was observed in specific oxygen consumption by 1°C decrease in the temperature. Between 6 and 37°C, the effect of temperature on oxygen consumption rate could be described using an Arrhenius model, i.e., $q_{O_2} = q_{O_2}^0 \cdot e^{-E/RT}$. The activation energy, E , in this equation was similar for different cells (between 80 and 90 kJ/mol), indicating the action of a similar mechanism for the effect of temperature on oxygen consumption. © 1999 John Wiley & Sons, Inc. *Biotechnol Bioeng* 64: 349–356, 1999.

Keywords: oxygen consumption; respiration; cell culture; temperature; cell density

INTRODUCTION

Oxygen consumption rate (OCR) is an important parameter for cell culture fermentation since it relates directly to the total viable cell concentration in the bioreactor:

$$OCR = q_{O_2} \cdot X_v, \quad (1)$$

where q_{O_2} is the specific oxygen consumption rate of the cells, and X_v is the viable cell concentration in the bioreactor. The OCR can be measured by several techniques, and the information obtained can be used for implementation of automatic feeding strategies for optimal and consistent production. A feeding strategy based directly on OCR was

illustrated for batch and fed-batch operation (Zhou and Hu, 1994). Similarly in a perfusion system, the feed rate to the bioreactor can be altered in response to variations in OCR (Kyung et al., 1994; Ozturk et al., 1995; Ozturk, 1996). Utilization of OCR for optimizing insect cell baculovirus systems was illustrated in the literature (Qualitz and Kwong, 1994).

Consistent and stable operation in perfusion bioreactors can be achieved by keeping the feed or dilution rate (D , in volumes per day) proportional to viable cell density (X_v , in million cells/mL). Under these conditions a constant cell-specific perfusion rate (CSPR), defined as $CSPR = D/X_v$, is established for the bioreactor operation. We have demonstrated that operating the bioreactor at constant CSPR allows for the operation of high-density reactors at a defined medium composition and ensures a constant physiological state for the cells. CSPR control allows stable bioreactor operation for consistent and optimal production (Ozturk et al., 1995). The OCR can successfully be used for the estimation of viable cell density (X_v) in the bioreactor for CSPR control (Ozturk, 1996; Ozturk et al., 1996).

There are several methods of determining the oxygen consumption rate in the bioreactor. In dynamic methods, the gas delivery is altered temporarily and the response in dissolved oxygen (DO) is analyzed for the determination of OCR (Kyung et al., 1994; Merten et al., 1986). In the continuous methods, the aeration system is not disturbed and the OCR can be calculated from mass balances using gas inlet and outlet composition. Together with the gas flow rate, the difference in gas composition allows on-line determination of the OCR (Bonarius et al., 1995; Eyer et al., 1995). The OCR can also be measured using gas phase oxygen composition (measured by a gas analyzer) and dissolved oxygen (measured by DO probe) using a mass transfer method (Ramirez and Mutharasan, 1990; Eyer et al., 1995). We have also illustrated a simple yet powerful technique for “real-time” OCR measurements using a DO controller (Ozturk et al., 1995). The gas flow rates to the bioreactor and the DO probe reading can be used to estimate the OCR via a mathematical model. These calculations are done on a computer and the OCR is estimated on-line with no disturbance to the system.

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A valuable measurement technique for OCR involves the use of a respirometer. Traditionally, the respirometer is used for off-line determination of oxygen consumption rate. Cells are removed from the bioreactor and placed into a well-agitated chamber. The air is then removed from the system (hence no mass transfer), and the DO in the chamber is monitored against time. The DO decreases as a result of oxygen consumption by the cells. The data collected for DO are analyzed to determine the OCR. It should be mentioned that the respirometer can also be operated in situ. In this case the cells are circulated through a well-agitated chamber containing a DO electrode. When the circulation is interrupted momentarily, the DO decreases due to oxygen consumption and the OCR can be calculated from the DO-time plot. The use of a respirometer is tedious, and its utility is not suitable for process control purposes since a respirometer does not give a continuous signal. However, the respirometer can be used to verify the oxygen consumption rate obtained from more complicated methods. It is also a useful tool for studying the effect of various parameters on OCR.

In this study we utilized the respirometer to measure cell-specific oxygen consumption rates of different cell lines. Some issues related to DO probe response and the analysis of data for OCR determination were studied using a mathematical model. The use of respirometer resulted in quick and accurate determination of specific oxygen consumption of cells and its influence by culture conditions. The effects of cell density and temperature on the oxygen consumption rate were investigated for different cell lines over a wide range of variables.

THEORY

When a suspension of cells is placed in a well-agitated chamber and the gas exchange to the cells is eliminated, the dissolved oxygen concentration decreases due to consumption of oxygen by the cells:

$$-\frac{dC_L}{dt} = q_{O_2} \cdot X_v = \text{OCR}, \quad (2)$$

where C_L is the dissolved oxygen concentration in the medium, t is time, q_{O_2} is the cell-specific oxygen consumption rate, and X_v is the cell density.

The DO electrode measures the dissolved oxygen concentration, and the response of the probe is described as:

$$\tau_p \frac{dC_p}{dt} = C_L - C_p, \quad (3)$$

where C_p is the probe reading, and τ_p is the probe response time. The probe response time is the time constant for probe measurement, and it is a measure of how fast the probe equilibrates with the oxygen concentration exposed. The probe response time for the DO electrode is directly related to probe type, geometry, membrane, and electrolyte properties. A thinner membrane, for instance, results in a faster

probe response with lower probe response time. DO electrodes can become sluggish (large probe response time) when the same membrane is used for a long time (aging).

The solution to Eqs. (2) and (3) results in the following model, relating the DO response to oxygen consumption rate:

$$C_p = C_p^o + \text{OCR} \cdot \tau_p (e^{-t/\tau_p} - 1) - \text{OCR} \cdot t. \quad (4)$$

When the probe reading (C_p) is monitored against time, a nonlinear data regression can be used to extract the OCR. The success in regression is enhanced if the probe response time (τ_p) is measured independently. This can be easily done by inserting the DO probe from one oxygen concentration to another in a step change mode and analyzing the data on the electrode reading against time. Eq. (3) can then be used to estimate the probe response time. As mentioned above the probe response time depends on several factors and usually varies between 20 and 90 s (Commercial Ingold Probe, Mettler Toledo Application Note, 1992). For cases of low OCR and/or fast probes, the probe response time can be neglected and an "approximate" model for the OCR can be used:

$$C_p = C_p^o - \text{OCR} \cdot t. \quad (5)$$

Thus a linear model can be used to fit the data and the OCR is obtained from the slope of probe reading data against time.

MATERIALS AND METHODS

OCR Measurement Using a Respirometer

The respirometer was obtained from Yellow Springs Instruments (Yellow Springs, OH). The respirometer utilizes oxygen microprobes (YSI Model 5357) with an adapter threaded into a plunger (Model 5093). A water-jacketed bath accommodates three (3) glass chambers (working volume 3–7 mL) where cell suspensions are stirred with glass-coated magnetic bars. In the experimental setup, the DO electrodes were connected to an amplifier and the data was recorded on a computer using LabView data acquisition software (National Instruments, Austin, TX). About 5 mL of cell suspension was placed into each glass chamber and allowed to equilibrate to the desired temperature with agitation at approximately 700 rpm for about 10 min. The agitation did not cause any damage to the cells during measurement, which lasted in some cases up to 20–30 min, and cell viability as measured by dye exclusion method remained high (>95%). Then agitation was stopped, the plungers were carefully inserted into the glass chambers, and all the air was removed. The agitation was immediately started and the DO response was recorded. The data collected by LabView data acquisition system were analyzed by Kaleidagraph software on a Macintosh computer and the OCR was extracted using a nonlinear regression based on full analysis (Eq. (4)).

The DO probes were calibrated at 100% using air saturated water in the chambers and at 0% using nitrogen flushed in a rubber glove. The probe response time was obtained when the probe was inserted into air saturated water after the zero calibration. The response time varied between 20 and 40 s.

Cell Lines, Source, and Medium

Three different cell lines were utilized in this study: baby hamster kidney (BHK), murine hybridoma, and Chinese hamster ovary (CHO) cells. In each case, the cells were obtained from 15-L bioreactors at cell densities around 20 million cells/mL. All the cells were cultivated routinely in continuously fed perfusion bioreactors (stirred tank reactor with cell retention) in serum-free DMEM:F12-based media. Temperature and pH were maintained at 36.5°C and 50% DO, respectively.

Effect of Cell Density

Cell suspensions at 20 million cells/mL were removed from the bioreactors aseptically in 50-mL centrifuge tubes. Cells were spun down at 1000 rpm for 10 min, and the supernatants were separated. The cells were then diluted in different amounts of supernatants to achieve cell concentrations between 1 and 20 million cells/mL. The use of supernatant as a diluent maintained a constant media composition, so the only variable changing in this experiment was the cell density.

Effect of Temperature

To study the effect of temperature on specific oxygen consumption rate, cell suspensions were first diluted to about 4 million cells/mL. The use of lower cell density (4 million cells instead of 20 million cells/mL) caused the OCR to decrease and prolonged the monitoring time of DO, allowing for the accumulation of more data points during the experiment. The water bath temperature for the respirometer was varied between 6 and 37°C. Cells were incubated in the measuring chamber for up to 10 min for temperature equilibration. Once the desired temperature was reached in the cell suspension, the oxygen consumption rates were evaluated as a function of temperature as described above. The effect of temperature was studied for BHK, murine hybridoma, and CHO cell lines.

RESULTS AND DISCUSSION

OCR Measurement Using a Respirometer

Figure 1 shows the time profiles of dissolved oxygen during oxygen consumption rate measurements for CHO cells at a density of 20 million cells/mL. The experiment was performed at 37°C. The dissolved oxygen started decreasing

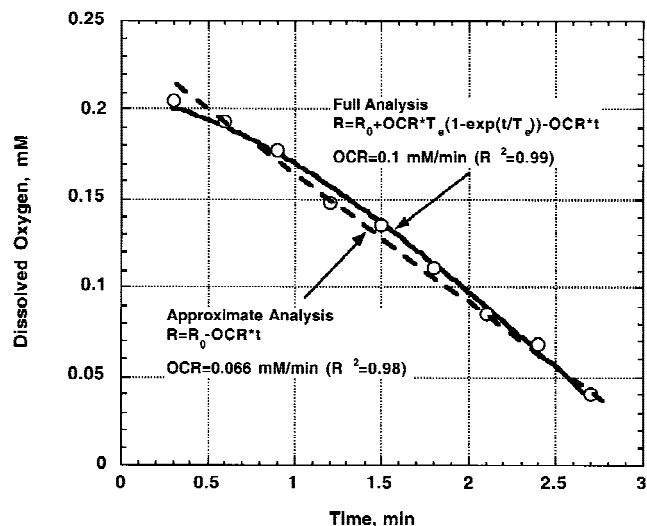


Figure 1. Time profile of dissolved oxygen during oxygen consumption rate measurements for CHO cells at a density of 20 million cells/mL (37°C). Experimental data are analyzed for oxygen consumption rate using full (solid line) and approximate (dashed line) analysis.

from air saturation (0.21 mM dissolved oxygen at 100% air saturation), and it dropped to zero in about 3 min. The data collected were used in a nonlinear regression analysis to extract the OCR using Eq. (4) (full analysis). The model with an electrode response time of 30 s yielded an oxygen consumption rate of 6 mM/h.

Figure 1 also depicts the results of an approximate analysis where electrode response time is neglected. The approximate analysis requires a linear fit to the data with a slope equal to OCR. The apparent OCR was then calculated to be 5 mM/h. Thus, the error in neglecting the electrode response time was about 20%. The error in this case was small as the electrode response time was relatively short (30 s). With other probes, the error can be significant as explained below.

Significance of Using Full Analysis in Respirometer Experiments

As mentioned earlier, the probe response time should be included in the analysis of data from respirometers. Although the error can be minimal for low oxygen consumption rates and for fast probes, the error in neglecting the probe response time on OCR determination can be significant. The problem lies in overlapping of the probe time constant with the time constant of experimental data collection.

In order to access the error in neglecting the probe response time in DO response, we differentiate Eq. (4) and obtain the following equation:

$$-\frac{dC_p}{dt} = \text{OCR}_{\text{app}} = \text{OCR} \cdot (1 - e^{-t/\tau_p}). \quad (6)$$

This equation relates the actual and apparent (with no electrode response time) oxygen consumption rates. The error is

directly related to time constant for the probe (τ_p). Using the time constant for the measurement ($\tau_O = C_p^0/\text{OCR}$) and dimensionless time ($\tau = t/\tau_O$), we obtain the following equation to relate actual OCR to apparent OCR:

$$\text{OCR}_{\text{app}} = \text{OCR} \cdot (1 - e^{-\alpha\tau}), \quad (7)$$

where $\alpha = \tau_O/\tau_p = C_p^0/(\text{OCR} \cdot \tau_p)$ is a dimensionless constant.

It should be noted that the approximate analysis uses a linear fit to the data points and results in an average OCR for the measurement. It is usual practice to neglect the first portion of the data points, as that is regarded as the lag phase. Thus between $x < \tau < 1$, where x is the first time point included in the analysis, the approximate analysis yielded an "apparent" OCR which is related to the actual OCR by

$$\overline{\text{OCR}}_{\text{app}} = \text{OCR} \cdot \left[\int_x^1 (1 - e^{-\alpha\tau}) d\tau / \int_x^1 d\tau \right]. \quad (8)$$

Using the above analysis, case studies were evaluated in which the actual OCR was varied and the effect of probe response time was investigated.

Figure 2 compares the actual and the apparent OCR values for different probe response times. The apparent OCR is always lower than the actual rate, and the difference increases with probe response time. The difference is negligible at low OCR values. As is clear from Eq. (8) the effect of OCR and the electrode response time can be combined as a single parameter, $\alpha = \tau_O/\tau_p = C_p^0/(\text{OCR} \cdot \tau_p)$. This parameter is the ratio of time constants for oxygen consumption and electrode response. To minimize the effect a separation of time constants is needed. A value of α greater than 5 results in 99% accuracy for OCR determination. For lower values of α , the error can be significant.

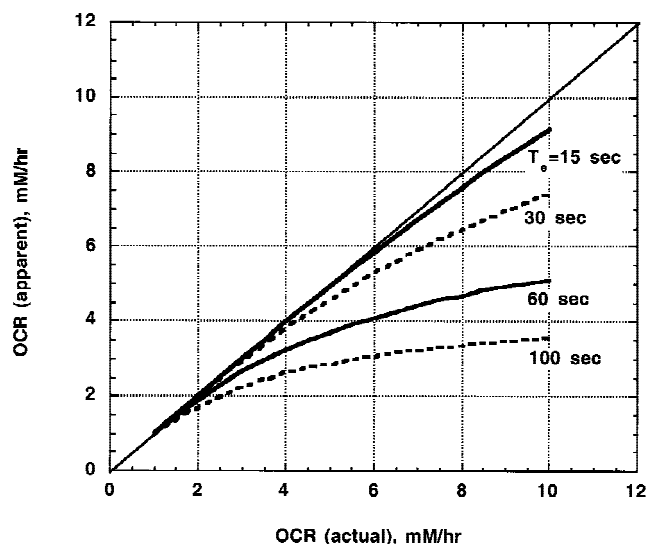


Figure 2. Effect of probe response time on the measurement of oxygen consumption rate. Neglecting the probe response time (approximate analysis) results in apparent oxygen consumption rates lower than actual ones.

Effect of Cell Density

When the cell density was varied between 1 and 20 million cells/mL, the oxygen consumption rate increased linearly (data not presented). From the oxygen consumption rates measured and the known cell concentrations, the specific oxygen consumption rates were calculated. Figure 3 presents the specific oxygen consumption rates for CHO cells at 37°C. The specific oxygen consumption did not change with the cell density in these experiments. Since the specific oxygen consumption rate is not dependent on cell density, the OCR can be used directly to estimate viable cell density.

The relationship between specific oxygen consumption rate and cell density is a controversial one, as evident from the literature. Most research demonstrates the absence of a relationship between cell density and specific oxygen consumption rate (Fleischaker and Sinskey, 1981; Ozturk et al., 1995; Qualitz et al., 1993; Yamada et al., 1994) between 0.2 and 20 million cells/mL. However, some research showed a strong cell density effect on oxygen consumption (Danes and Paul, 1961; Sand et al., 1977; Wohlpert et al., 1990). When the cell density was altered between 0.05 and 0.6 million cells/mL, the specific oxygen consumption rate was not altered (Danes and Paul, 1961). However, cell concentrations above 0.6 million cells/mL resulted in a decrease in specific oxygen consumption. Sand et al. (1977) reported a 10-fold decrease in specific oxygen consumption rate when cell density was increased from 0.001 to 10 million cells/mL for lymphocyte cultures. Finally, Wohlpert et al. (1990) observed a 4-fold decrease in cell specific oxygen consumption by an increase in cell density from 0.6 to 10 million cells/mL for hybridoma cells.

It should be pointed out that some of the cell density effect data presented in the literature are generated during

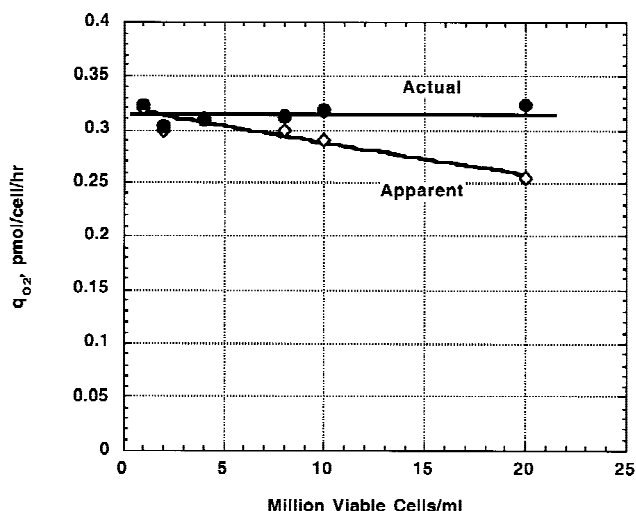


Figure 3. Specific oxygen consumption rates for CHO cells at 37°C obtained at different cell densities. The apparent specific rates decreased with increasing cell densities. This trend is not real and is a result of using approximate analysis. When the full analysis is used, the actual specific oxygen consumption showed no dependence on cell density.

the batch or fed-batch cultures. Under these conditions the cell density changes so as the media composition and physiological state of the cells. The batch cultures, for instance, start with low cell density and high nutrient levels. High cell concentrations are obtained at the late phase of exponential growth with low nutrient levels. Cell growth rate is also higher at the beginning (low cell density) and lower at the end (high cell density) of the culture. It is known that cells are metabolically more active at the active growth stage, and the cell density effect in these cases may actually be due to growth rate variations. In studying the cell density effect it is desirable (1) to independently alter the cell density and (2) to keep all other variables such as media composition, culture age, and cell growth rate constant. In this study, cell density was altered independently by diluting the cells from a concentrated sample. This strategy was followed by Wohlpart et al. (1990) and Yamada et al. (1994). It is interesting to note that Yamada et al. (1994) also generated data using the cell density data directly, and the specific oxygen consumption rate was not dependent on cell density for four different cell lines used. In the second set, cell density changed as a function of culture time and there was a cell density effect. It should be pointed out that cellular environment was not maintained constant in these studies as evident from culture viability and perfusion rate reported (Yamada et al., 1994).

Our conclusion from the review of literature is that, although the cell density effect can be real as presented in the literature, artifacts due to experimental technique and data analysis can be responsible for some of the trends reported. Our analysis presented here indicates the significance of probe time constant and the need for its inclusion in the data analysis for oxygen consumption rate measurements. When approximate analysis instead of full analysis was used, the apparent specific rates obtained in our experiments also demonstrated a decreasing trend with cell density as depicted in Fig. 3. This artificial trend is created by neglecting the probe time constant (30 s) from the analysis. When full analysis is used, the actual specific oxygen consumption showed no dependence on cell density.

As mentioned above, utilization of cell specific perfusion rate to control the perfusion rate allows for the operation of high-density reactors at a defined medium composition and ensures a constant physiological state for the cells (Ozturk, 1996; Ozturk et al., 1996). Under these conditions the cell density effect can be extracted from the bioreactor data, as the only variable changing in the bioreactor is the cell density. Oxygen consumption rates (OCR) were measured on-line for a CSPR-controlled bioreactor cultivating BHK cells at 36.5°C. Cell densities are determined daily from the bioreactor samples. When the OCR values are plotted against viable cell concentrations, a linear correlation is established (Fig. 4) with a slope equal to specific oxygen consumption rate. As clearly shown from Fig. 4, the slope of the line does not change with the cell density, indicating the absence of any cell density effect when cell density was varied between 0.6 and 16.5 million cells/mL. The cell-specific oxygen

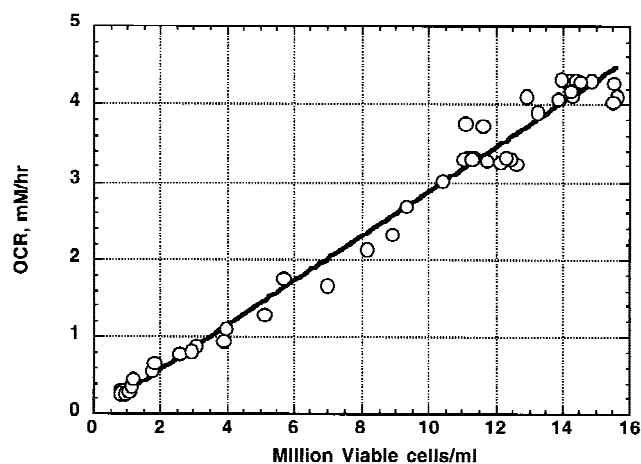


Figure 4. Oxygen consumption rates measured during the cultivation of BHK cells as a function of cell densities determined daily from the bioreactor samples. Data were obtained in a well-controlled perfusion bioreactor operated at 36.5°C. The slope of the line presented yields a constant specific oxygen consumption rate of $q_{O_2} = 0.29$ pmol/cell/h.

consumption for BHK cells at 36.5°C can be established as 0.29 pmol/cell/h from Fig. 4. We have previously studied glucose consumption and lactate production from hybridoma cells and came to the same conclusion (Ozturk et al., 1997). The cell density did not affect the specific metabolic rates for glucose and lactate in the range of cell densities studied (2–14 million cells/mL) for a well-controlled perfusion bioreactor.

Oxygen consumption rate can be used for the estimation of viable cell concentrations. The success of such a prediction lies on how constant the specific oxygen consumption rate is. In the exponential phase of batch culture, the specific oxygen consumption rate was observed to be constant and cell number in the bioreactor was estimated with good precision (Qualitz et al., 1993). The specific oxygen consumption rate is known to decrease in the late exponential and decline phase of the batch culture (Ramirez and Mutharasan, 1990; Zhou and Hu, 1994) and the success of prediction is expected to be lower for extended batch or fed-batch operations. In the perfusion culture of BHK cells with CSPR control, we obtained a constant specific oxygen consumption rate. Under these conditions we could estimate viable cell concentrations in the bioreactor successfully (Fig. 5).

Effect of Cell Line

Of the cell lines used in this study, the murine hybridoma cell line had the lowest specific oxygen consumption rate (Table I). At 37°C the specific consumption rate was determined to be 0.2 pmol/cell/h. The BHK and CHO cell lines had about 50% higher oxygen consumption, and at this temperature they had oxygen consumption rates equal to 0.28 and 0.31 pmol/cell/h, respectively. Note that the off-line value obtained for the BHK cell line using the respirometer (0.28 pmol/cell/h) is very close to the one obtained on-line in the bioreactor (0.29 pmol/cell/h, obtained from Fig. 4). It should be noted that cells were exposed to the

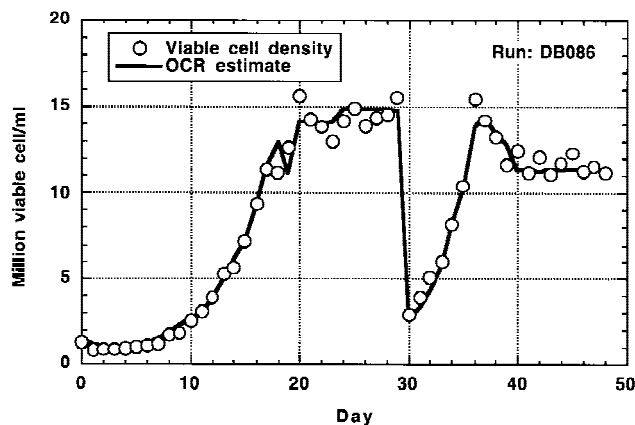


Figure 5. Estimation of viable cell concentration using oxygen consumption rates during the cultivation of BHK cells at 36.5°C. Experimental data (circles) and prediction from OCR measurement (solid line) are presented.

same medium composition during the measurements in both bioreactor and respirometer experiments. The specific oxygen consumption rates for mammalian cells vary between 0.05 and 0.5 pmol/cell/h (Eyer et al., 1995; Fleischaker and Sinskey, 1981), and the measured specific oxygen consumption rates were within this range.

Effect of Temperature

The effect of temperature on specific oxygen consumption rate has not been well studied in the literature for the mammalian cells. The specific oxygen consumption is expected to be lower at low temperatures. Studies on the effect of temperature on the oxygen consumption rate are limited to a narrow range of temperatures. This is mainly due to the fact that the oxygen consumption rate is normally studied under growth conditions. Mammalian cells require temperature ranges for optimal growth between 33 and 38°C (Pugh, 1988; Kretzmer et al., 1997; Chuppa et al., 1997). Under these conditions a 10–20% decrease in the specific oxygen consumption rate was observed when temperature was reduced by 1°C for hybridoma and BHK cells (Pugh, 1988; Kretzmer et al., 1997; Chuppa et al., 1997). When the cell density is limited by the oxygen transfer rate, operating at a lower temperature may increase the cell number maintained in the reactor and thus yield a higher bioreactor productivity.

We have studied the effect of temperature on specific oxygen consumption rates between 6 and 37°C. The effect of temperature on cell specific oxygen consumption rate is presented in Figs. 6, 7, and 8, for CHO, murine hybridoma, and BHK cells, respectively. As expected, the specific oxygen consumption decreased at lower temperatures. Temperature influenced the specific oxygen consumption rate similarly for the cells studied, and about 10% decrease was observed in specific oxygen consumption by reducing the temperature by 1°C. This result confirms the data obtained from previous studies (Pugh, 1988; Kretzmer et al., 1997;

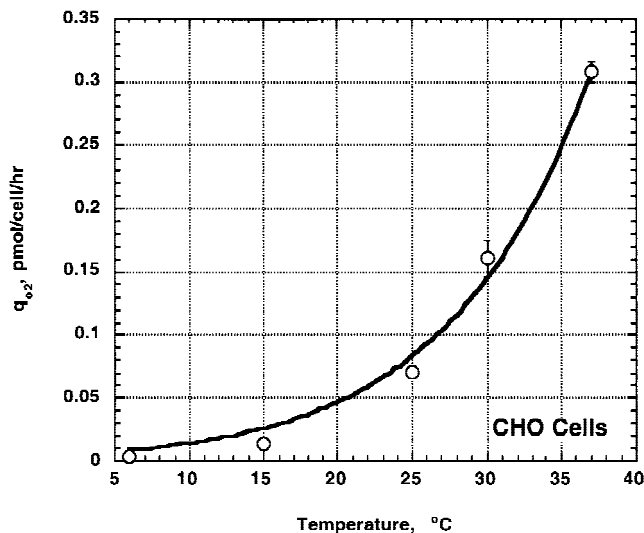


Figure 6. Effect of temperature on CHO cell-specific oxygen consumption rate.

Chuppa et al., 1997). Below 15°C, the oxygen consumption was one order of magnitude lower than what is obtained at 37°C.

The influence of temperature on specific oxygen consumption rate followed similar trends for the three cell lines used in this study. The effect of temperature was observed to be exponential in all cases. Temperature dependency of rate constants (chemical or enzymetical) is traditionally modeled using the Arrhenius model (Johnson et al., 1954). This model can also be applied to a biological process such as oxygen consumption rate. Under these conditions, the data from examination of the temperature effect can be fitted to the following rate equation:

$$q_{O_2} = q_{O_2}^0 \cdot e^{-E/RT}, \quad (9)$$

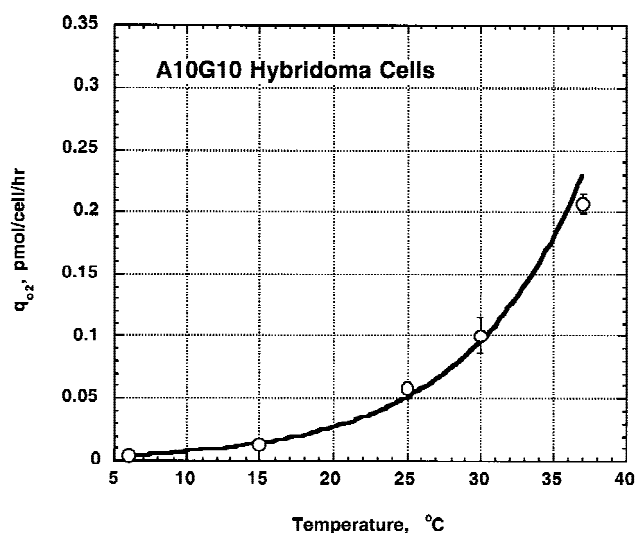


Figure 7. Effect of temperature on murine hybridoma cell-specific oxygen consumption rate.

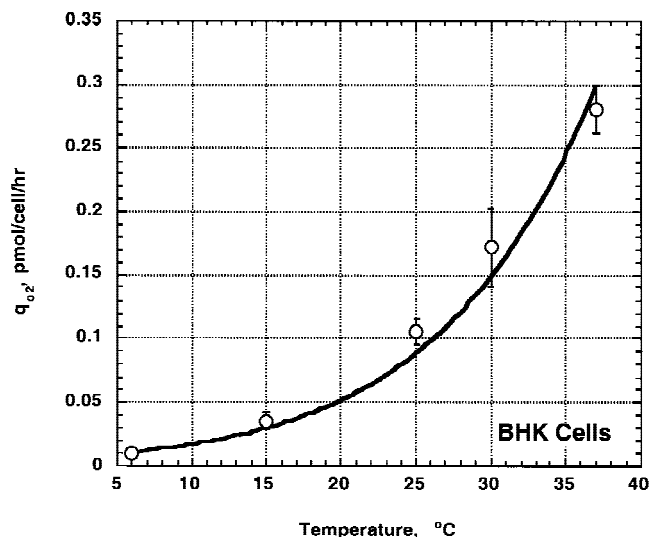


Figure 8. Effect of temperature on BHK cell-specific oxygen consumption rate.

where E is the activation energy, R is the ideal gas constant, and T is the absolute temperature in degrees K. Figure 9 presents the prediction of this model for the temperature dependency of specific oxygen consumption rate. For all the cell lines used, Eq. (9) successfully described the effect of temperature on specific oxygen consumption rate with Arrhenius constants presented in Table I. The activation energy, E , in this equation was similar for different cell lines indicating a similar mechanism for the effect of temperature on oxygen consumption. It should be pointed out that an activation energy of 100 kJ/mol was reported in the literature for the BHK cell line by studying the oxygen consumption rate between 34 and 37°C (Kretzmer et al., 1997). It is interesting to note that the activation energy we obtained is

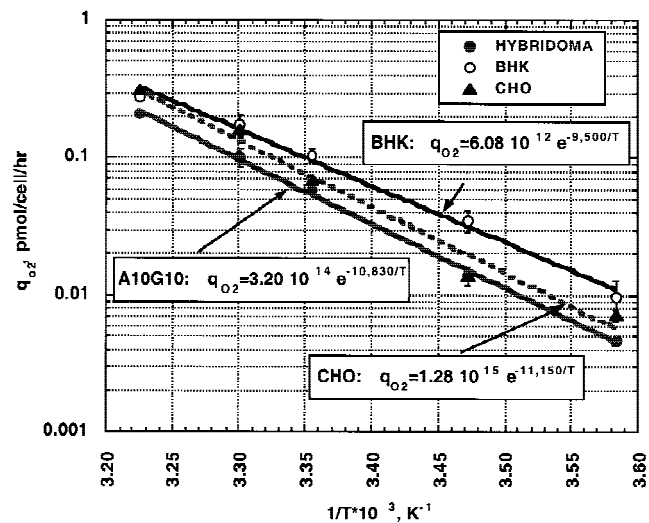


Figure 9. Effect of temperature on different cell lines. The temperature dependency of oxygen consumption rate could be modeled using an Arrhenius model. The activation energies for oxygen consumption rate are very similar (80–90 kJ/mol) for the cell lines studied.

Table I. Arrhenius constants for different cell lines used in this study.

Cell line	q_{O_2} at 37°C (pmol/cell/h)	$q_{O_2}^0$ (pmol/cell/h)	E (kJ/mol)
BHK	0.30	0.61	79
Hybridoma	0.22	32	90
CHO	0.31	128	93

very similar to those of enzymatic reactions (Johnson et al., 1954). Oxygen is used in the electron transport chain in a complicated process. It is possible that one of the key enzymes is limiting the process and that we are dealing with the activation energy of this particular enzyme.

As mentioned above, the specific oxygen consumption rate is cell specific. This may necessitate the establishment of temperature effect for every cell line. However, we demonstrated that activation energy for oxygen consumption is similar for the cell lines used in this study. Thus one can estimate the influence of temperature on specific oxygen consumption rate by Eq. (9) with an activation energy (E) of 80–90 kJ/mol for the cell line considered. Only activation energy and oxygen consumption rate measurement at a single temperature point are needed to establish Eq. (9) and to estimate the effect of temperature.

CONCLUSIONS

Several conclusions can be drawn from the results of this study:

1. The respirometer is a useful tool for measuring the oxygen consumption rate. The measurements can be performed in minutes with an accurate determination of oxygen consumption. It allows to study one variable at a time as illustrated here for the effects of cell density and temperature. Data measured in an off-line respirometer is comparable to those measured on line as demonstrated here for BHK cell line as long as cells are maintained under the same conditions.
2. Neglecting probe response time for the calculation of oxygen consumption rate can result in errors, especially when the probe response is slow and/or the oxygen consumption is fast.
3. The specific oxygen consumption rate is not affected by the cell concentration for cell densities between 1 and 20 million cells/mL.
4. The specific oxygen consumption is cell line dependent. For a given temperature, the murine hybridoma cells had the lowest consumption and the CHO cells had the highest consumption.
5. Temperature influenced the specific oxygen consumption rate similarly for the cells studied and about 10% decrease was observed in specific oxygen consumption by reducing the temperature by 1°C. The temperature dependence of specific oxygen consumption could be expressed in an Arrhenius form, i.e., $q_{O_2} = q_{O_2}^0 \cdot e^{-E/RT}$.

The activation energy, E , in this equation was similar for different cell lines (between 80 and 90 kJ/mol), indicating a similar mechanism for the effect of temperature on oxygen consumption.

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